PRODUCT REQUIREMENTS DOCUMENT & SRS SPECIFICATION

Peer-to-Peer (P2P) Micro-Lending Platform

Document Version	1.0.0
Document Type	PRD & Software Requirements Specification (SRS)
Target Architecture	Web Application, Mobile Apps (iOS / Android) and RESTful APIs
Loan Range	BDT 5,000 - 50,000 (short-term micro-credit)
Status	Draft for Review

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1. Executive Summary & Objective

The primary objective of this platform is to bridge the gap between unbanked or underbanked individuals requiring fast, short-term micro-credit (BDT 5,000 to 50,000) and retail investors seeking yield on idle capital. By eliminating traditional banking intermediaries, the platform lowers operational overhead, reduces interest rate spreads, and provides automated, transparent risk assessment.

Core Problems Solved

- **Borrower Accessibility:** Eradicates paperwork delays and strict collateral requirements for micro-loans.
- **Lender Yield Generation:** Offers individual investors higher returns compared to conventional retail savings accounts.
- **Trust & Security:** Enforces identity authentication, legal compliance, and automated escrow management.

2. Key Personas & Role Matrix

Borrower

- Submits official identification documents for Know Your Customer (KYC) verification.
- Creates micro-loan applications defining loan amount, purpose, and payback period.
- Withdraws approved funds to third-party accounts (Bank / Mobile Financial Services).
- Repays scheduled monthly or weekly installments (Equated Monthly Installments - EMI).

Lender (Investor)

- Deposits fiat funds into the platform's multi-currency digital wallet.
- Assesses loan applications on the marketplace using risk scores, yield percentages, and tenure.
- Commits funds toward single or syndicated micro-loans.
- Receives interest and principal repayments back into the internal wallet.

Admin / Risk & Compliance Officer

- Audits and approves or rejects KYC identity packages.
- Resolves transaction disputes and monitors platform liquidity metrics.
- Enforces Anti-Money Laundering (AML) flags and handles default recoveries.

3. System Architecture & Workflows

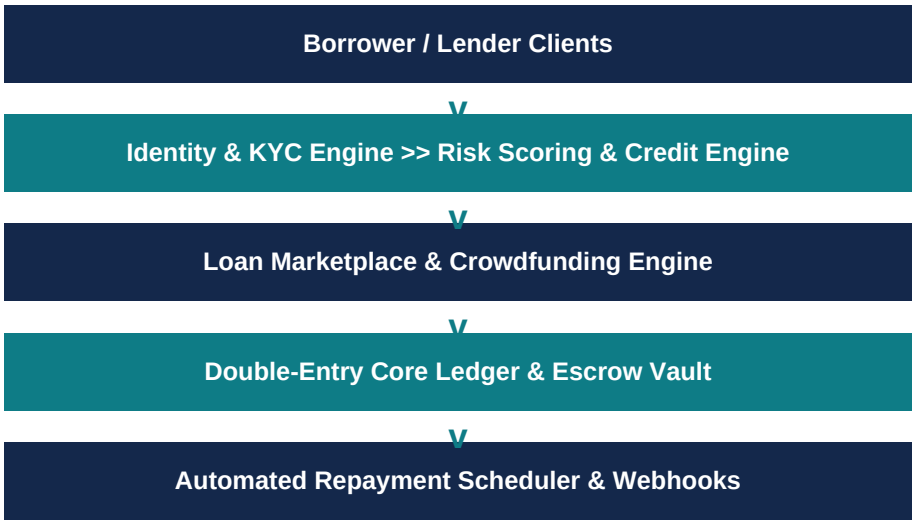

Figure 1: High-level system component flow

Core Architectural Mechanisms

- **Double-Entry Bookkeeping:** All balance mutations follow strict ledger accounting (`Assets = Liabilities + Equity`). Every financial movement consists of at least one debit and one credit entry to prevent currency duplication and phantom balances.
- **Escrow Holding Mechanism:** Funds committed by lenders are locked in a dedicated escrow account (`ESCROW_LOCKED`). Capital is only released to the borrower's spendable balance once the loan listing reaches 100% funding. If the funding window expires, funds return to the lender's wallet instantly.
- **Automated Repayment Engine:** A background job scheduler executes installment deductions on specified due dates, applying automated penalty rules for overdue accounts.

4. Epics, User Stories, and Acceptance Criteria

Epic 1: User Onboarding & Identity Verification (KYC)

Category	Functional Requirement - Compliance & Security
Priority	Critical (Must Have)

User Story 1.1: Document & Biometric Submission

> **As a newly registered user (Borrower or Lender),**
>
> **I want to** submit my National ID (NID) and a live facial capture,
>
> **So that** my real-world identity is verified to unlock deposit, withdrawal, and lending privileges.

Acceptance Criteria (Gherkin Format)

```
Scenario: Successful submission of identification documents
  Given the user is authenticated and possesses the status "KYC_UNVERIFIED"
  When the user uploads high-resolution front and back images of their National ID
  And successfully completes the real-time liveness biometric scan
  Then the system updates the account status to "KYC_PENDING"
  And blocks any financial transaction attempts until approved.

Scenario: Rejection of unsupported or oversized files
  Given the user is on the KYC document upload page
  When the user submits a file format other than PNG, JPG, or PDF
  Or the file payload size exceeds 5 MB
  Then the system displays an inline validation error:
      "File format must be PNG, JPG, or PDF under 5MB"
  And prevents network payload transmission.
```

Epic 2: Core Digital Wallet & Double-Entry Ledger

Category	Functional Requirement - Ledger & Funds Management
Priority	Critical (Must Have)

User Story 2.1: Wallet Top-Up via External Payment Gateway

> **As a verified lender,**
>
> **I want to** deposit money into my platform wallet via bank transfer or Mobile Financial Services,
>
> **So that** I have available balance to allocate toward micro-loans.

Acceptance Criteria (Gherkin Format)

```
Scenario: Successful deposit credit to user wallet
  Given the user has the status "KYC_VERIFIED"
  When the user initiates a deposit of 10,000 BDT via the integrated
      payment gateway
  And the gateway returns a synchronous 200 HTTP success webhook
  Then the system records a Credit transaction of 10,000 BDT to the
      User Balance account
  And records a Debit transaction of 10,000 BDT to the Platform
      Gateway Clearing account
  And updates the visible available balance in real time.

Scenario: Handling duplicate webhook events (Idempotency)
  Given a deposit transaction has already been processed and credited
  When the payment gateway triggers a duplicate webhook callback with
      the same transaction reference
  Then the ledger engine identifies the duplicate idempotent key
  And rejects redundant balance adjustment, logging a
      "DUPLICATE_EVENT_IGNORED" audit notice.
```

Epic 3: Loan Marketplace & Crowdfunded Escrow

Category	Functional Requirement - Marketplace & Order Matching
Priority	High (Must Have)

User Story 3.1: Loan Request Creation

> **As an eligible borrower,**
>
> **I want to** list a loan request defining the principal amount, tenure, and purpose,
>
> **So that** retail lenders can evaluate and finance my application.

Acceptance Criteria (Gherkin Format)

```
Scenario: Eligible borrower lists a new loan request
  Given the borrower holds a "KYC_VERIFIED" status
  And has zero active overdue loans on the platform
  When the borrower submits a loan application of 20,000 BDT with a
      3-month tenure
  Then the system generates a new listing with the status
      "MARKETPLACE_LISTED"
  And displays the calculated repayment schedule and interest rates
      to prospective lenders.
```

User Story 3.2: Loan Funding & Escrow Disbursement

> **As an active lender,**
>
> **I want to** invest funds into a listed loan,
>
> **So that** my committed capital is held in escrow until fully funded and subsequently disbursed.

Acceptance Criteria (Gherkin Format)

```
Scenario: 100% funding achieved and loan disbursed
  Given a loan listing is active with 5,000 BDT remaining to reach
      the target amount
  When a lender commits 5,000 BDT from their available wallet balance
  Then the system locks the 5,000 BDT under the status "ESCROW_LOCKED"
  And detects that the target funding threshold (100%) has been achieved
  And transitions the loan state to "DISBURSED"
  And transfers the total principal into the borrower's spendable
      wallet balance.
```

Epic 4: Repayment Engine & Default Management

Category	Functional Requirement - Billing & Debt Collection
Priority	High (Must Have)

User Story 4.1: Automated Recurring Installment Deduction

> **As the automated billing platform,**
>
> **I want to** execute scheduled installment deductions on loan due dates,
>
> **So that** lenders receive their principal and interest distributions automatically.

Acceptance Criteria (Gherkin Format)

```
Scenario: Successful automated installment collection
  Given an active loan with an upcoming installment due date matching today
  And the borrower's wallet balance is equal to or greater than the
      installment value
  When the daily repayment batch job runs
  Then the system debits the exact installment amount from the
      borrower's balance
  And credits proportional principal and interest shares to the
      respective lenders' balances
  And updates the installment status to "PAID".

Scenario: Insufficient funds trigger grace period and penalty
  Given the borrower's wallet balance is insufficient on the due date
  When the scheduled collection job encounters a failed debit attempt
  Then the loan enters a 3-day status of "GRACE_PERIOD"
  And alerts the borrower via SMS and push notification
  And initiates a daily late penalty surcharge of 0.5% after grace
      period expiration.
```

5. Non-Functional Requirements (NFR & SRS Constraints)

Dimension	Technical Specification
Data Protection & Encryption	All Personally Identifiable Information (PII) must be encrypted at rest using **AES-256**. All external communication requires **TLS 1.3**.
Transaction Concurrency	The matching engine must enforce database-level pessimistic locking on loan objects during funding calls to prevent race-condition overfunding.
Financial Auditability	Database ledger tables must be append-only. Balance updates must never execute via manual overwrite queries (`UPDATE balance`), only via signed transaction records.
System Availability	The core API infrastructure must guarantee **99.9% uptime** with high-availability multi-zone deployment.